



GAËL RÖTHLIN

FULL-STACK DEVELOPER | COMPUTER SCIENCE STUDENT (BUT)

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Passionate about new technologies and the design of computerized systems. I analyze objects and architectures in depth to better understand them, then rebuild them with my own means. Strongly focused on full-stack development, distributed systems and local AI.

EXPERIENCE

INTERNSHIP: INNOVATIVE DIGITAL & LOW-TECH

Jan. – Apr. 2026

University of Franche-Comté, CROUS BFC

- Developed a **local, low-tech AI solution** (no data sent to the cloud).
- Built a **3D object viewer in VR**.
- Provided technical support to the CROUS BFC.

INTERNSHIP: FULL-STACK DEVELOPER (EUPHRON PROJECT)

Apr. – May 2025

University of Franche-Comté, Claude Oytana Learning Centre

- Developed a **web/mobile application** to facilitate sports-equipment donations between students.
- Worked across the full front-end / back-end stack.

TECHNICAL PROJECTS

Sagittarius: Local AI platform

- Web platform for local AI via **Ollama**, with no data sent to the cloud.
- Dedicated workspaces (Code, Mail, Chat), **Mistral / LLaMA** support, real-time response streaming.

Vue 3 • NodeJS • Ollama • CSS

Smart-Waste: Distributed IoT system

- Smart urban-waste management: **multi-threaded TCP** server in Java, **ESP32** sensors, MongoDB database, geolocated reporting module.
- Containerized with Docker, tested using **BDD** (Cucumber / JUnit).

Java • ESP32 • MongoDB • Docker • Cucumber

SpotCLI: Terminal Spotify client

- Spotify client with a **terminal UI (TUI)** built from scratch in C, using the Spotify Web API.

C • Spotify API • TUI

B.I.E.N.: Digital awareness

- Educational, interactive website on the **environmental impact of digital technology** and the CO2 it generates.

Web • Awareness

Roi des Roses: Board game

- Board game in Java using the **MVC** paradigm and the Boardifier framework, as a team project (5 students).

Java • MVC • Boardifier

DosOok: Sound-based transmission

- Java program transmitting text through an **encrypted sound signal**.

Java • Signal processing

DotIC: Halftone art generator

- Turns images into artworks made of dots whose size varies with luminance.

Python • PIL • Halftone

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript, PHP, SQL

Back-end: NodeJS, SQL, PHP, multi-threaded TCP servers

Front-end: Vue.js (Vue 3), JavaScript, CSS, TUI

AI & Data: Ollama, Mistral / LLaMA, local AI, MongoDB

Tools & DevOps: Git, Docker, Linux, Cucumber / JUnit (BDD)

Embedded / IoT: ESP32, microcontrollers, VR

KEY STRENGTHS & EDUCATION

Key strengths

- Writing and managing specifications and databases.
- Installing & configuring equipment (servers, PCs).
- Software updates and security patching.

Education

BUT in Computer Science

2023 – present

IUT Nord Franche-Comté, University of Franche-Comté

Technological Baccalauréat STI2D, with Honours

2023

SIN specialization, Lycée Edouard Belin

Languages: French (native), English (C1, advanced)